

- ii. Task prioritization, due dates, and status tracking.
- iii. Automated workflow routing, reminders, and escalation mechanisms.
- iv. Real-time task dashboards for individuals and leaders.

E. Paper Preparation & Multi-level Approval Workflows

- i. Collaborative drafting environment with track changes and version comparison.
- ii. Multi-stage approval workflow including but not limited to:
Member → Group Leader → Delegation Leader → IC Division → CS/NA/CSS → DG → MoPSW (where applicable).
- iii. Locking and finalization of approved documents to prevent unauthorized edits.

F. Correspondence Group Management

- i. Nomination and management of Correspondence Group members.
- ii. Upload and review of correspondence group tasks, inputs, and progress updates.
- iii. Group-level consolidation and reporting mechanisms.

G. Meeting-time Collaboration & Discussion Interface

- i. Controlled real-time collaboration interface during live meetings.
- ii. Discussion boards for agenda-wise deliberations and last-minute changes.
- iii. Role-based editing and commenting permissions during meetings.

H. Dashboards, Reports & Analytics

- i. Role-based dashboards (Admin, Delegation Leader, Coordinator, Member).
- ii. Committee-wise, meeting-wise, and agenda-wise dashboards.
- iii. Tracking of tasks, submissions, pending approvals, and outcomes.
- iv. Analytical insights on participation, performance, and timelines.
- v. Downloadable and configurable reports in XML, Excel, and PDF formats.

I. Calendar, Alerts & Notifications

- i. Integrated calendar displaying upcoming meetings, deadlines, and milestones.
- ii. Automated alerts and notifications for agenda updates, task assignments, approvals, and meetings.
- iii. Notification delivery through in-portal alerts and email.

J. User Management & Role-Based Access Control

- i. Centralized administrative console for user and role management.
- ii. Assignment of users to committees, working groups, and roles.
- iii. Role-based access enforcement across all modules and data.
- iv. Comprehensive access logs and audit trails for all user actions.

K. Compliance & Quality Requirements

- i. The application shall strictly adhere to the approved FRS/SRS and all applicable Government of India usability, accessibility, security, and performance standards.
- ii. Any deviations shall require prior written approval from DGS.
- iii. The SI shall ensure that the solution is extensible, maintainable, and future-ready.

Detailed Functional Requirements are mentioned in Annexure 2

3.4 Technology Stack

The System Integrator (SI) shall propose, implement, and maintain an **open, scalable, secure, and future-ready technology stack** for the IMO Strategic Digital Platform. The proposed stack shall be subject to approval by the Directorate General of Shipping (DGS) and shall comply with **Government of India digital, security, accessibility, and interoperability standards**.

The technology stack shall be based on **open standards and industry-accepted frameworks**, ensuring interoperability, ease of maintenance, and long-term sustainability. The architecture shall support **horizontal and vertical scalability** to handle increased users, data volumes, and future functional expansion. Security shall be embedded across all layers of the application, and the system shall be designed to avoid **vendor lock-in**, allowing DGS to migrate, extend, or maintain the solution independently in the future.

The frontend shall be a **web-based, responsive interface**, developed using standard frameworks (such as Angular, React, or equivalent), compatible with all major modern browsers and devices (desktop, tablet, and mobile). The solution shall comply with Government usability and accessibility guidelines, including **WCAG 2.1**.

The backend shall be developed using **enterprise-grade, open frameworks**, following a service-oriented or microservices-based architecture. It shall support modular development, stateless services, centralized business logic, workflow management, and role-based access enforcement. All functionalities shall be exposed through **well-documented APIs** to enable future integrations.

The integration layer shall follow an **API-first approach** using RESTful services and open data formats. Secure API management mechanisms shall be implemented for authentication, authorization, monitoring, and future extensibility. The data layer shall use **enterprise-grade databases** suitable for both structured and unstructured data, with appropriate indexing, encryption, automated backups, and recovery mechanisms to ensure data integrity and availability. The platform shall be hosted exclusively on a **MeitY-empanelled Cloud Service Provider**, with appropriate cloud services for high availability, scalability, backup, disaster recovery, and monitoring. Cloud infrastructure and application security controls shall comply with **ISO 27001, ISO 27017, ISO 27018, and CERT-In guidelines**. Industry-standard tools shall be used for development, testing, version control, and CI/CD, with separate environments for development, testing/UAT, production, and disaster recovery.

The **entire source code and related artefacts** shall be the exclusive property of DGS. The SI shall ensure that no proprietary dependencies are introduced that restrict future modification or transfer of the system. All third-party components used shall be properly licensed and disclosed.

Any deviation or change in the approved technology stack shall require **prior written approval from DGS**, and non-compliance with approved standards shall be treated as a contractual breach.